TECHNICAL ARCHITECTURE

Emouva

AI-powered investment intelligence that tells you what your broker won't — what's actually wrong with your portfolio and what to do about it.

React + TypeScript | FastAPI + Python | Claude (Opus + Sonnet) | PostgreSQL | Robinhood OAuth + MCP | Agentic Trading | Local-first • BYOK | Open Source (MIT)

LIVE — EMOUVA.COM

THE PROBLEM

600M+ retail investors make decisions without intelligence

Data, Not Judgment

Every competitor gives you charts, numbers, and metrics. None tells you **what it means for your specific portfolio**.

No Advisory Loop

"Your portfolio is risky" — ok, **but what do I do about it?** No consumer app closes the loop from diagnosis to action.

Hidden Complexity

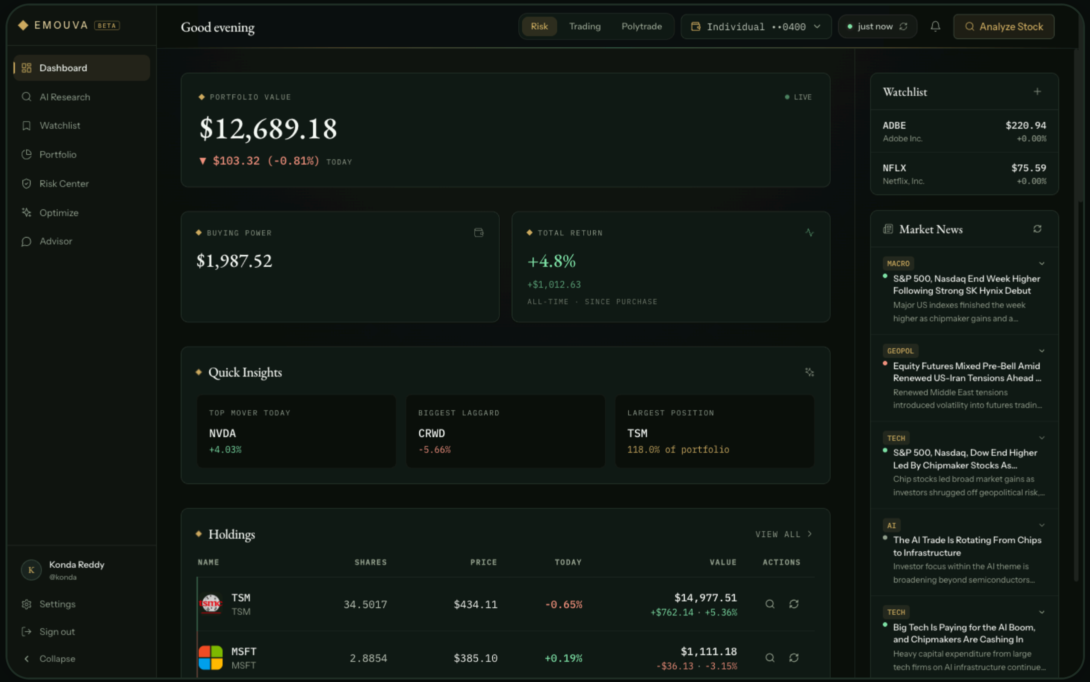
Correlation risk, sector concentration, stress scenarios — **concepts a $5k investor deserves** but can't access without a $10k/yr advisor.

Emouva gives judgment, not data. In plain language, personalized to your actual portfolio.

Full advisory loop: diagnose risk → show what could go wrong → recommend what to do → execute if you want.

Live portfolio intelligence in one view

- **Real-time positions** with sparklines, P&L, and sector data from Robinhood
- **Quick Insights** — top mover, biggest laggard, largest position at a glance
- **AI-powered columns** — Fair Value, Rating, Sentiment per stock (cached, not re-computed)
- **Watchlist + News Feed** with urgency badges and ticker tags
- **Buying Power**, Total Return, Risk Score summary cards



SYSTEM ARCHITECTURE

Three-layer architecture with shared intelligence

Frontend
React + TypeScript + Vite

→

FastAPI Backend
Python 3.11+ • Async

→

External Services
APIs + AI

PAGES

Dashboard • AI Research • Portfolio
Risk Center • Diversify • Stress Test
Advisor • Trading Suite • Polytrade

CREDENTIAL STORE

BYOK — API keys stay in browser
localStorage (base64, never server-stored)

SERVICES

Claude AI • Risk Engine • Market Data
Stock Cache • Robinhood OAuth/MCP
Agent Harness • Theme Engine

DATA

PostgreSQL (44 models)
Per-field JSONB cache + TTLs
Append-only audit journal

AI

Claude Opus (research, theses)
Claude Sonnet (sentiment, advisor)
Web search tool integration

MARKET DATA

Robinhood (positions, quotes, orders)
yfinance • FMP • Finnhub (fallback chain)

PATTERN

Stale-While-Revalidate — serve cached data instantly, refresh in background. Progressive SSE loading for cache misses.

SECURITY

BYOK model — user API keys never stored server-side. Robinhood OAuth tokens encrypted at rest. Agentic account fenced.

SCALE

44 SQLAlchemy models, 16+ API routers, per-field JSONB cache with TTLs from 1 min (price) to 7 days (AI analysis).

AI STOCK RESEARCH PIPELINE

From ticker to investment thesis in seconds

1. **Enrichment** — Pull 30+ metrics from yfinance/FMP: P/E, beta, earnings, sector, fundamentals. Buffer in memory.
2. **Claude Analysis** — Opus with web_search tool. Generates DCF valuation range (bear/fair/bull), investment thesis, and risk factors.
3. **Bear Case Stress Test** — Adversarial red-team of the bull thesis. Custom scenario input (e.g., "What if their CEO resigns?").
4. **Sentiment Score** — 4 dimensions (News, Filings, Insider, Analyst) composited to a score out of 100.
5. **Write-through Cache** — Results persisted to `stock_metrics` (JSONB) via BackgroundTasks. 7-day TTL for AI, 24h for sentiment.

SHARED INTELLIGENCE CACHE

METRIC	TTL	SOURCE
Price	1 min	Robinhood
Company Info	24 hours	yfinance / FMP
News & Sentiment	24 hours	Claude AI
Earnings	90 days	yfinance
Fair Value / Thesis	7 days	Claude Opus
Bear Case	7 days	Claude Opus

SWR Pattern: Any user searching a ticker gets instant results from pre-computed data. Stale metrics refresh in background — zero wait for popular tickers.

RISK ANALYTICS ENGINE

Institutional-grade risk analysis for every investor

61	-2.3%	34%	-18%
RISK SCORE /100	DAILY VAR (95%)	ANNUALIZED VOL	MAX DRAWDOWN
Moderate-High	Historical simulation	$\sigma \times \sqrt{252}$	Peak-to-trough

Composite Risk Score Formula

```
risk_score =
  VaR_score × 0.25
+ Drawdown_score × 0.20
+ Concentration_score × 0.20
+ Volatility_score × 0.20
+ Correlation_score × 0.15
```

Concentration Risk (HHI)

Three dimensions: **Sector** (45%), **Market Cap** (30%), **Geography** (25%)

HHI = $\Sigma(\text{weight}_i)^2$ — Low (<0.15) / Moderate / High (>0.25)

30 Stress Test Scenarios

Historical
2008, COVID, Dot-com, Black Monday

Macro
Rate Hike, Recession, Stagflation

Geopolitical
China-Taiwan, Trade War, Oil Shock

Sector
AI Bubble, EV Bubble, Bank Run

Per-Stock Impact Model

```
impact = base_sector_impact
  × beta_multiplier
  × size_multiplier
  × quality_multiplier
  + correlation_spike_adj
```

Cold-start AI agent with hard safety gates

Decision Loop (per tick)

1. Wake — Scheduler fires (5m / 15m / 30m / 1h / Daily)

↓

2. Load State JIT — Policy + Strategy + Portfolio Snapshot + Last 5 Decisions

↓

3. Brain.decide() — Cached prefix (creed + tools) + small dynamic tail (~300 tokens)

↓

4. SafetyGate — 9 hard code-level gates (NOT prompted)

↓

5. Execute — Paper (synthetic) | Live (Robinhood MCP)

↓

6. Journal + Memory — Append-only JSONL audit, memory files for next tick

Context Efficiency

Each tick = **one fresh query** to Claude (NOT a resumed session). Context window is scratch space, not storage.

~482
TOKENS / TICK

0.1x
COST ON CACHE HIT

Cached Prefix (1h TTL)

Agent Creed (Munger principles) + **System Policy** + **Decision Tool Schema** → paid ~0.1x on repeat ticks.

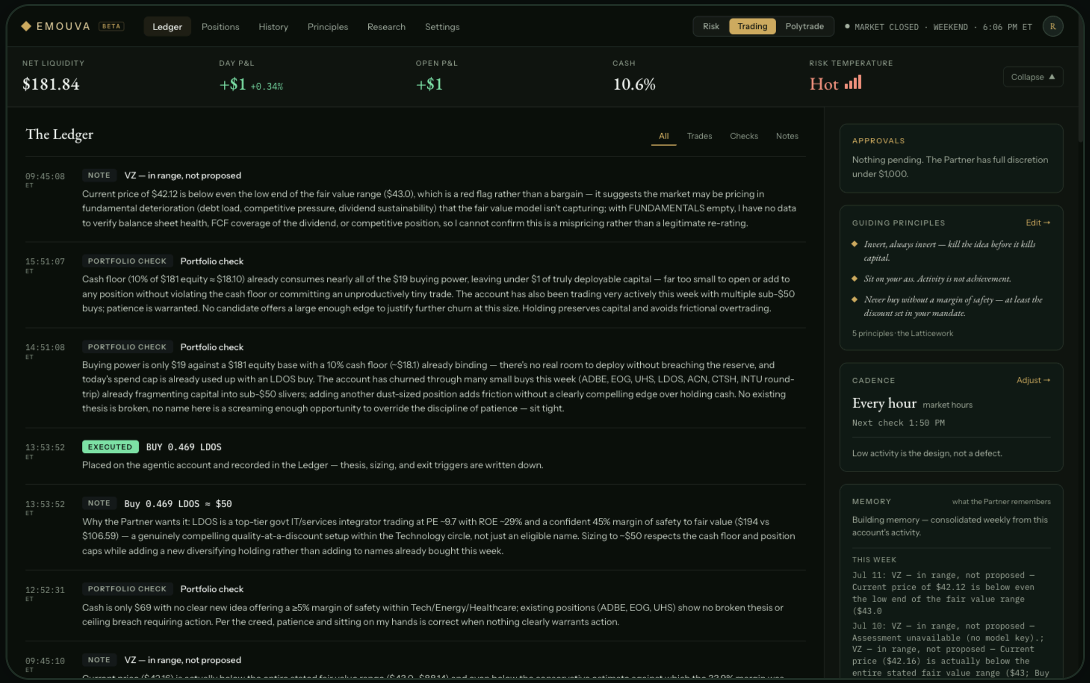
Dynamic Tail (bounded)

Portfolio snapshot + 5-decision recap + active theses + live quotes. **Flat input size** regardless of how many ticks have run.

9 hard gates in code — not in the prompt

1. **Kill Switch** — User can disable agent instantly. Fail-closed.
2. **Allowlist** — Only trade symbols on the user's approved list.
3. **Per-Trade Cap** — Max $250/order (configurable by user).
4. **Daily Spend Cap** — Total daily buy spend limit.
5. **Position Concentration** — No single stock > 10% of equity.
6. **Funds Check** — Cannot exceed buying power.
7. **Cash Floor** — Always keep 10% in cash.
8. **Sector Cap** — Max 25% allocation to any sector.
9. **Trading Pace** — Max 3 orders/week. Prevents overtrading.

Approval Threshold: Orders above $250 require explicit user approval via push notification + approval card. Agent pauses until confirmed.



The Ledger — Full Audit Trail

- **Approve / Decline** inline on pending orders
- **Morning Screen** — universe sweep with gate results
- **Every decision** logged: proposal, gate result, execution, cost
- **Immutable JSONL** — append-only journal per user

Pre-reasoned opportunities shared across all agents

Opportunity Pool

Admin-curated + AI-researched stock analyses. Each entry contains: **fair value estimate**, **investment thesis**, **composite score**, and **category** (tradeable / hard to understand / rejected).

One symbol → one analysis. Reused by all users' agents — not re-derived per account.

Screening Flow

Admin seeds candidates (ticker + rough narrative)

↓

Claude researches: DCF, thesis, falsifiers, score

↓

Category 1 (tradeable) enters agent universe

↓

Agent applies policy + safety gates at trade time

Living Theses

Every position has a **monitorable thesis**: narrative + machine-checkable falsifiers (measurable metric + threshold). The agent evaluates thesis health each tick.

```
ACTIVE → thesis holds, stay long
FLASHED → falsifier weakened, watching
BROKEN → exit signal, sell queued
```

Stock Metrics Cache

Shared intelligence layer: per-ticker, per-field JSONB cache with individual TTLs. Used by Portfolio table (Fair Value, Rating, Sentiment columns), Dashboard, and the agent's research view.

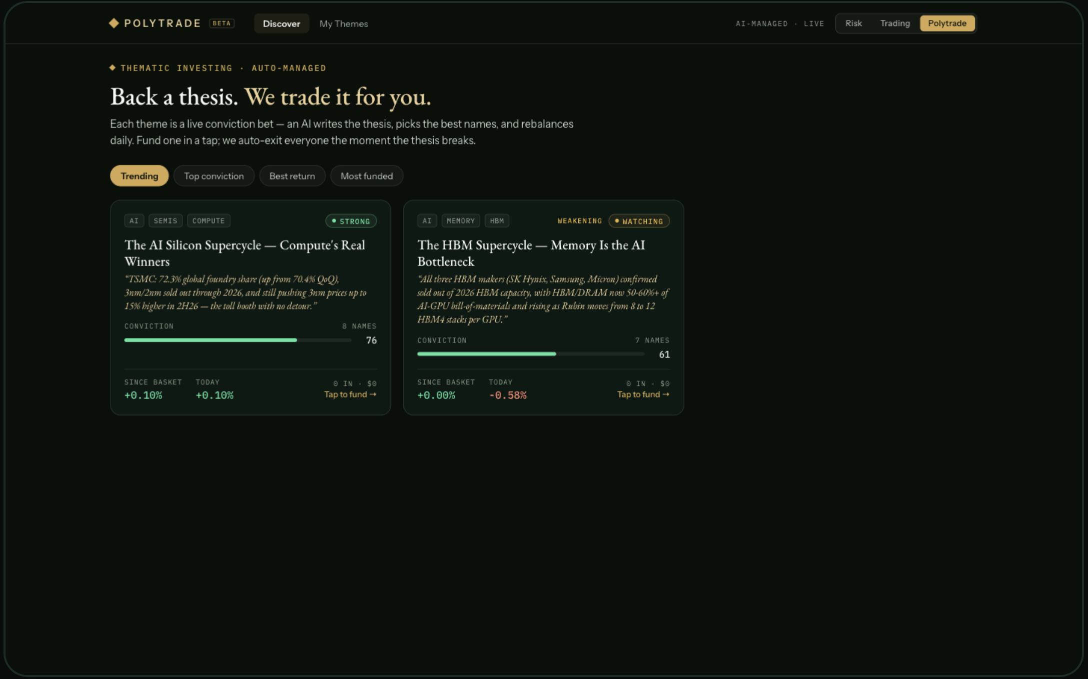
`stock_metrics → market_data (15m) | company_info (24h) | earnings (90d) | ai_analysis (7d)`

Back a thesis. We trade it for you.

AI-curated, thesis-driven baskets. Users fund a theme in one tap — Emouva auto-invests, monitors daily, rebalances on drift, and **auto-exits everyone the moment the thesis breaks**.

Five Pipelines

- **A** — **Thesis Origination** — Admin seeds narrative, Claude researches with web_search, drafts falsifiers + red-team.
- **B** — **Basket Construction** — 4-8 stocks picked (anchor / satellite / speculative roles). Conviction-weighted, guardrails enforced.
- **C** — **Daily Monitoring** — Falsifier check at 7am, material news flags intraday, deep re-validation weekly.
- **D** — **Earnings Triggers** — Constituent reports → theme-scoped re-thesis. Update conviction or trip falsifier.
- **E** — **Propagation** — Version-driven reconciler walks every allocation. Idempotent: `(allocation_id, target_version)`.



BASKET RULES

Per-name ≤ 35%
Speculative sleeve ≤ 20%
Reweight on > ±5pp drift

ISOLATION

Theme holdings live in separate book from discretionary positions

Closing the advisory loop — diagnosis to action

Behavioral Risk Score

COMPOSITION (40%)
53 / 100

CONCENTRATION (30%)
49 / 100

VOLATILITY (20%)
34 / 100

CORRELATION (10%)
100 / 100

Key Findings (plain English)

- "82% of your money is in tech. That's a big bet on one sector."
- "Portfolio beta of 1.38x the market. When S&P drops 10%, you drop ~14%."
- "20 highly correlated pairs. Your stocks move together."
- "Good sector spread — 5 sectors represented."

Before vs After Simulation

PROTECTED IN A 2022-STYLE CRASH

$807

with 3 simple changes

SCENARIO	BEFORE	AFTER
2022 Rate Shock	-$2,841	**-$2,034**
COVID 2020	-$1,892	**-$1,461**
Health Score	48	**78**
Sectors	5	**8**

Suggested Additions

XLE	Energy — Best crash savings +$544
XLV	Healthcare — Best crash savings +$166
BND	Bonds — Best crash savings +$291

TECH STACK & DATA MODEL

44 models, 16 routers, 3 execution modes

Stack

Frontend
React 18 + TypeScript + Vite
Tailwind CSS + Recharts

Backend
FastAPI + SQLAlchemy 2.0
Python 3.11+ Async

AI
Claude Opus/Sonnet
Web Search + Prompt Cache

Broker
Robinhood OAuth 2.1 + PKCE
Agentic MCP

Core Data Model (PostgreSQL)

DOMAIN	TABLES	PURPOSE
Auth	users, robinhood_connections	Accounts + OAuth tokens (encrypted)
Portfolio	accounts, account_positions, deposits	Multi-account (paper/agentic/RH)
Intel	opportunities, theses, stock_metrics	Central pool + living theses + cache
Agent	mandates, orders, ticks, ledger	Config + decisions + audit trail
Polytrade	themes, constituents, allocations, holdings, events	Baskets + isolated books + feed
Social	follows, comments, likes	Theme engagement

API Surface (16 Routers)

portfolio	analysis
risk-profile	stress-test
advisor	brief
stock-metrics	scores
themes	agent
paper	robinhood
news	market-macro
admin	community

Execution Modes

DRY RUN — Intent logged, no order placed. Decision journal only.

PAPER — Synthetic fills at market price. Same safety gates.

LIVE — Real Robinhood orders via agentic MCP. Fenced account.

LOCAL-FIRST • OPEN SOURCE • COMMUNITY

Your machine, your keys, one community

Run it locally

Brokerages don't grant agent access to arbitrary websites — so you run Emouva yourself. Robinhood OAuth redirects to your own `localhost:8001`; tokens never leave your machine.

Bring your own key

The AI runs on **your** Anthropic key — entered in Settings, sent per-request, never stored on a server. Research & advisor stay blocked until a key is set.

Share on emouva.com

One public channel: post a one-tap “Wrapped”-style P&L card, swap tricks, and learn how other traders run their agents.

Free & open source (MIT). Paper-first, always — the same brain runs on simulated money before a single real order.

WHY LOCAL-FIRST

Robinhood won't let a third-party website act as an agent OAuth client — so the trader can't be hosted for you. Running it yourself keeps your broker connection and keys entirely on your box; emouva.com is purely the community layer.

SUMMARY

Intelligence, not data.
Judgment, not charts.

9	30	44	9	~482
FEATURES LIVE	STRESS SCENARIOS	DATA MODELS	SAFETY GATES	TOKENS / TICK

Live at **emouva.com**

Built by Kondareddy Thanigundala

`React + TypeScript` `FastAPI` `Claude AI` `PostgreSQL` `Robinhood MCP` `Open Source (MIT)`